

Sequence similarity without overclaiming

What does a good alignment tell us, and what does it not tell us?

By the end, you should be able to...



Distinguish sequence identity, similarity and homology.



Score a short alignment and interpret a local match.



Read a BLAST hit using coverage, identity and E-value together.

Reconnect with the biology

DNA sequences and protein sequences use different alphabets;
compare like with like.

Think silently -> explain to a neighbour -> check the measurement.

Similarity is observed; homology is inferred

Measured similarity

Evolutionary relationship

Identity counts identical aligned symbols.

Similarity can use a scoring scheme that rewards related amino acids.

Homology means common ancestry; do not report 70% homology.

Global and local answer different questions

Global: full sequences

ACGTTAC
ACGCTAC

Local: selected region

... GTT ...
... GTT ...

Global alignment compares full sequences under a scoring scheme.

Local alignment finds a high-scoring matching segment.

A short local match may cover only a small fraction of a protein.

A scoring rule must be stated

**score = 2 x matches -
mismatches - 2 x gaps**

State the units and the assumptions.

For our toy DNA alignment: match +2, mismatch -1, gap -2 per position.

Scores from different rules are not directly interchangeable.

A biological interpretation comes after the arithmetic.

One exact local-alignment step

$$H[i,j] = \max(0, \\ H[i-1,j-1] + \text{substitution}, \\ H[i-1,j] - 2, H[i,j-1] - 2)$$

Each cell considers a match/mismatch, a gap from either direction, or zero.

Zero permits a local alignment to start fresh.

The maximum is exact for this recurrence and scoring model, not proof of a biological mechanism.

Why search tools use shortcuts

01

A database may contain many sequences.

>

02

BLAST uses seed-and-extend heuristics to find useful local matches efficiently.

>

03

A heuristic may miss matches; its output still needs an audit.

Read three columns, not one

Hit	Identity	Coverage	E-value
A	95%	10%	0.02
B	70%	90%	1e-30

Identity: how alike are aligned positions?

Query coverage: how much of the query is represented?

E-value: expected chance matches at least this strong under the search model.

The E-value depends on the search space

For a comparable alignment score, a larger search space can increase the E-value.

An E-value is not the chance the functional annotation is wrong.

Also inspect low complexity, domains, identifiers and database provenance.

Three approaches can be combined

Algorithmic guarantee

Empirical performance

Exact optimization solves a specified mathematical problem.

Heuristics trade exhaustive search for speed.

Learned representations depend on training data; a tool can contain all three.

Worked example: score a short alignment

ACGT aligned to ACCT, with match +2 and mismatch -1.

Predict the result and one limitation before calculating.

Work through the reasoning

Positions 1, 2 and 4 match: $3 \times 2 = 6$.

Position 3 mismatches: subtract 1.

Total score = 5; identity = $3 / 4 = 75\%$.

Pause: make a defensible claim

Hit A has 95% identity across 10% of a protein; hit B has 70% identity across 90%. Which is the stronger starting point for a full-length relationship, and what still needs checking?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

B covers far more of the query. Inspect domain architecture, alignment quality and provenance; neither hit alone proves identical function.

Today's practical

W04: score a fixed alignment, fill a tiny local-alignment table using a supplied function, and audit two hypothetical search hits.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

1 alignment

2 identity

3 homology

4 heuristic

5 E-value

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: inspect affine gap penalties and substitution matrices; deriving log-odds is not examined.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S06 NCBI: How BLAST Works and BLAST Statistics: The Expect Value.

S26 NCBI: How BLAST Works (2022 workshop).

S28 Smith and Waterman (1981). Identification of common molecular subsequences. *Journal of Molecular Biology* 147:195-197.